

HELSINGBORG, Sweden

WMO: 026110

Lat: 56.030N

Lon: 12.765E

Elev: 144

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	18.7	8.9	8.7	16.9	13.1	10.8	20.9	25.2	40.3	22.6	40.6	5.7	40	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	79.6	65.9	76.5	64.4	73.5	63.0	68.0	75.7	66.1	73.2	64.5	71.2	7.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	93.6	71.1	63.4	87.7	69.4	61.7	82.6	67.7	32.3	75.6	30.8	72.9	29.6	71.2	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.5	15.7	DB	8.1	83.7	6.9	2.9	3.1	85.8	-0.9	87.6	-4.7	89.2	-9.7	91.3	(4)
(5)			WB	7.4	70.9	6.8	2.1	2.6	72.4	-1.4	73.6	-5.2	74.8	-10.2	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	32.9	33.5	37.4	45.5	53.3	59.0	63.2	63.2	57.0	48.6	41.2	35.9		
	DBStd	12.39	7.24	6.62	6.35	5.89	5.81	4.66	4.44	4.25	4.58	5.63	5.97	7.18		
	HDD50	2375	531	461	392	159	29	1	0	0	4	92	267	439		
	HDD65	6422	996	881	856	584	366	190	91	88	242	510	714	904		
	CDD50	1507	0	0	1	25	131	269	408	409	214	47	3	0		
	CDD65	79	0	0	0	0	3	9	34	31	2	0	0	0		
	CDH74	554	0	0	0	2	29	84	273	159	7	0	0	0		
	CDH80	63	0	0	0	0	2	8	42	11	0	0	0	0		
(14) Wind	WSAvg	7.8	8.4	8.6	8.1	7.9	7.5	7.3	6.8	7.0	7.5	7.9	8.2	8.6		
(15) Precipitation	PrecAvg	28.5	2.2	1.7	1.5	1.4	2.0	2.9	3.0	3.4	2.8	2.6	2.4	2.5		
	PrecMax	37.9	5.1	4.2	3.2	2.7	4.4	6.5	8.3	7.2	5.5	5.5	4.9	4.4		
	PrecMin	18.9	0.0	0.2	0.5	0.3	0.7	0.0	0.0	0.7	0.5	0.7	0.5	0.5		
	PrecStd	4.1	1.2	0.7	0.7	0.7	0.9	1.5	1.9	1.7	1.3	1.1	1.0	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	48.4	57.0	69.6	77.8	81.1	84.3	81.4	74.4	63.0	54.9	50.3		
		MCWB	46.2	46.0	47.9	56.6	62.6	66.0	67.4	66.4	63.4	57.2	53.4	48.9		
	2%	DB	45.4	45.8	51.7	63.7	72.7	76.3	80.5	78.3	70.3	59.6	52.7	48.4		
		MCWB	43.8	43.8	45.3	51.5	60.2	63.5	66.7	65.5	61.7	56.1	51.1	47.0		
	5%	DB	43.6	44.1	48.2	59.9	68.4	72.2	77.1	75.2	67.1	57.6	50.7	46.4		
		MCWB	42.2	42.3	43.8	50.5	57.4	61.5	65.0	64.3	60.2	54.7	49.0	45.1		
	10%	DB	42.1	42.3	45.6	56.0	64.4	68.5	73.4	72.0	64.4	55.9	49.0	44.7		
		MCWB	40.6	40.7	42.4	48.4	55.2	59.7	63.5	63.2	58.7	53.4	47.4	43.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	46.7	49.3	57.9	65.3	68.5	70.9	70.0	65.1	58.8	53.9	49.5		
		MCDB	47.6	47.6	53.9	67.2	74.1	76.9	79.8	77.4	71.3	60.9	54.8	50.1		
	2%	WB	44.2	44.2	46.4	53.7	61.8	64.9	68.2	67.5	62.8	56.7	51.3	47.1		
		MCDB	45.2	45.3	50.2	61.4	70.1	73.1	76.6	73.9	68.5	58.8	52.4	48.0		
	5%	WB	42.5	42.4	44.5	51.5	58.9	62.7	66.5	65.7	61.0	55.3	49.3	45.3		
		MCDB	43.4	43.7	47.4	58.1	65.9	70.2	74.1	72.0	65.7	57.1	50.4	46.2		
	10%	WB	40.7	40.7	42.7	49.1	56.3	60.8	64.7	64.2	59.5	53.7	47.6	43.5		
		MCDB	41.9	42.2	45.1	54.5	62.6	67.1	71.7	70.3	63.6	55.6	48.8	44.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	10.6	14.4	15.5	14.5	14.7	14.0	12.3	9.6	6.8	6.5		
		MCDBR	6.7	8.2	14.5	20.7	22.6	21.6	22.8	19.4	16.1	10.5	7.2	6.8		
	5% WB	MCWBR	6.7	6.7	9.9	12.1	12.6	11.4	10.9	9.4	9.4	7.6	6.6	6.7		
		MCWBR	6.5	7.6	12.8	18.3	19.5	19.0	18.9	16.3	14.5	9.8	6.9	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.332	0.354	0.387	0.384	0.378	0.400	0.400	0.369	0.353	0.333	0.300		
	taud		2.274	2.302	2.317	2.285	2.348	2.391	2.347	2.355	2.425	2.399	2.324	2.252		
	Ebn at Noon		187	230	255	263	272	275	265	257	249	221	176	158		
	Edh at Noon		17	24	30	36	36	35	36	34	27	22	16	14		
(44) All-Sky Solar Radiation	RadAvg		136	325	759	1291	1674	1769	1642	1321	932	452	172	92		
	RadStd		26	52	81	165	164	145	198	105	99	67	31	10		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	+38	(46)
(47)	Regional (64 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	-283	N/A	N/A	(47)

Nomenclature: See separate page